An Asymptotic Multiscale Symmetric Fusion Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*— Despite the high spectral resolution and abundant information of hyperspectral images (HSIs), their spatial resolution is relatively low due to limitations in sensor technology. Sensors often need to sacrifice some spatial resolution to ensure accurate light energy measurement when pursuing high spectral resolution. This tradeoff results in HSI's inability to capture fine spatial details, thereby limiting its application in scenarios requiring high-precision spatial information. HSI and multispectral image (MSI) fusion is a commonly used technique for generating high-resolution HSI (HR-HSI). However, many deep learning-based HSI-MSI fusion algorithms ignore correlation and multiscale information between input images. To address this issue, we propose an asymptotic multiscale symmetric fusion network (AMSF-Net) for hyperspectral and MSI fusion. AMSF-Net consists of two parts: the multilevel feature fusion (MFF) module and the progressive cross-scale spatial perception (PCP) module. The MFF module uses multistream feature extraction branches to perform information interaction between HSI and MSI at the same scale layer by layer, compensating for the spatial details lacking in HSI and the spectral details absent in MSI. The PCP module combines the input and output features of MFF, utilizes multiscale bidirectional strip convolution and deep convolution to further refine edge features, and reconstructs HR-HSI by learning the features of different expansion roll branches by connecting across scales. Comparative experiments with several state-of-the-art HSI-MSI fusion algorithms on four publicly available datasets, CAVE, Chikusei, Houston, and WorldView-3, are conducted to validate the effectiveness and superiority of AMSF-Net. On the Chikusei dataset, improvements were 9.1%, 12.5%, and 5.1%, respectively, on the indicators root-mean-square error (RMSE), error of relative global accuracy in synthesis (ERGAS), and spectral angle mapper (SAM), compared to the suboptimal method.**

***Index Terms*— Deep learning, hyperspectral and multispectral image (MSI) fusion, multilevel feature fusion (MFF).**

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I. Introduction

REMOTE sensing imaging serves as a crucial tool for obtaining information about ground objects. The hyperspectral image (HSI) acquired via remote sensing imaging technology comprises numerous consecutive narrow spectral bands spanning from visible to near-infrared wavelengths, making it indispensable for discriminating ground objects with comparable characteristics. Given the technological constraints of imaging devices, an individual sensor is unable to capture HSIs with concurrently high spatial and spectral resolutions, resulting in HSI often exhibiting excellent spectral resolution with limited spatial resolution. Nonetheless, numerous practical scenarios necessitate the integration of both high spectral and spatial resolutions, such as agricultural production [2], geological exploration [3], disaster detection [4], military surveillance [5], [6], and food safety [7]. In comparison to the properties of HSI, multispectral image (MSI) owns a superior spatial resolution while compromising on spectral resolution. In general, the low-resolution HSIs (LR-HSIs) and high-resolution MSIs (HR-MSIs) obtained by the sensor can be regarded as spatial and spectrally degraded versions of the HR-HSI, respectively. Thus, it is a necessary research direction to effectively fuse the spatial and spectral information of HR-MSI and LR-HSI to obtain HR-HSI.

HSI-MSI fusion can be divided into traditional fusion methods and deep learning-based fusion methods. Traditional fusion methods include HSI-MSI fusion methods based on pan sharpening [8], [9], [10], [11], HSI-MSI fusion methods based on matrix decomposition [12], [13], [14], and HSI-MSI fusion methods based on tensor representation [15], [16], [17], [18]. The fusion algorithm is based on pan sharpening to enhance the spatial resolution of MSI by incorporating details from a high-resolution panchromatic (PAN) image, thereby improving the resolution of MSI. Lu et al. [19] proposed a panchromatic sharpening method based on grayscale blending and spectral band adaptive detail fusion, which constructed an intensity blending model by establishing intensity and gradient constraints between the mixed-intensity image and the source

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image, and proposed a band adaptive detail fusion algorithm to fuse the details extracted from the mixed-intensity image with the details extracted by MSI in each band, achieving a good detail fusion effect. Although traditional models can achieve good fusion results, they rely on prior knowledge of manual design, which leads to high time costs. The representation ability of the model is limited, which makes it unsuitable for complex data distribution in real scenarios.

Unlike traditional model-based approaches, deep learning-based approaches enable feature extraction and abstraction by using an end-to-end deep network to learn the representations most relevant to the task. Therefore, the HSI-MSI fusion method based on convolutional neural network (CNN) has been widely used [20], [21]. For example, Liu et al. [22] proposed a remote sensing pan-sharpening method based on a two-stream fusion network (TFNet), which processes PAN and MSI images, respectively, to achieve effective feature extraction and fusion. Although this dual-stream design is efficient, it may still be limited by the processing power of its fixed two information sources in some complex scenarios.

Despite the good fusion outcomes achieved by the aforementioned deep learning-based fusion algorithms, there are still some unresolved issues. First, many deep learning-based algorithms for fusing HSI and MSI process two input source images separately, ignoring the correlation between the input images [23]. Second, as for the processing of image fusion algorithms on scale, some algorithms only fuse images at a single scale and do not fully consider the multiscale information of the input images, resulting in insufficient extraction of spectral and spatial information. However, some algorithms try to fuse at different scales to capture richer details and features. However, even so, these algorithms may still have shortcomings in cross-scale information integration and cross-modal information interaction. In addition, in some three-branch networks, although the fusion stream and the spatial–spectral dual stream are taken into account, there is still a lack of multilevel cross-modal information interaction, and most of the fusion algorithms based on deep learning do not fully consider the spatial characteristics of different dimensions of the image, which also makes the effect of image fusion need to be improved.

To solve the above problems, we propose a new three-branch fusion network for multispectral and HSI fusion, the asymptotic multiscale symmetric fusion network (AMSF-Net), which not only considers the spatial–spectral modal interaction but also makes full use of the multilayer fusion mechanism and cross-scale spatial perception strategy, so that the network can capture richer spatial context information. First, the input features of different scales are obtained by upsampling and downsampling the source images, and a multilevel feature fusion (MFF) module is constructed through the hyperspectral and MSIs of the same scale. The local spatial information and local spectral information are extracted from the two images, the fusion flow branch is carried out hierarchical fusion, and the spatial and spectral modal interaction is progressively carried out. To enhance the context aggregation ability of the fused features, inspired by [24], we combine the mainstream window partitioning strategy and residual structure, model the pixel interaction from the spatial and channel dimensions, and construct the dual-strategy self-attention block after fusion to explore the potential correlation of spatial channels. Second, to integrate the spatial–spectral fusion information at different scales, a progressive cross-scale spatial perception (PCP) module was proposed to learn the pyramid structure and reconstruct the HSI through step-by-step channel stitching and upsampling operations. To overcome the local receptive field limitations, we construct cross-scale connections to establish remote dependencies and extract multiscale nonlocal spatial information. The contribution of the methodology in this article can be summarized in the following three aspects.

1) We propose AMSF-Net for HSI-MSI image fusion, which includes MFF and PCP modules. It fuses spatial and spectral modes at multiple levels, combines features of different scales, and reconstructs spatial details.
2) The MFF module exploits spatial and spectral domain information, explores spatial-channel correlations, and fuses spatial–spectral information multistream and multilevel. It uses LFE blocks to avoid source image distortion.
3) The PCP module uses multiscale dilated convolutions and a bidirectional spatial perception (BSP) unit to extract and refine spatial features, enhancing the fused image quality.

The rest of this article is summarized as follows. Section II details the structure of the network constructed in this article. Section III presents the experimental results and analysis of the proposed method. Finally, the work of this article is summarized in Section IV.

II. Related Work

At the algorithm level, obtaining HR-HSIs by fusing two types of images is a promising research direction. In this section, we present some of the representative MSI and HSI fusion methods available.

In recent years, traditional methods have made many contributions to the convergence of HSI and MSI. Traditional fusion methods include fusion methods based on pan sharpening, fusion methods based on matrix factorization, and fusion methods based on tensor representation. For example, Guarino et al. [25] achieved unsupervised panchromatic sharpening by a principal component analysis. Considering the correlation between HSI and MSI, a fusion method based on matrix factorization [26] utilized the endmember matrix of HSI and the abundance map of MSI to regenerate a refined HR-HSI. Yokoya et al. [27] decomposed HR-MSI and LR-HSI into mixed pixels by coupled nonnegative matrix factorization (CNMF) and reconstructed high-quality HR-HSI using the abundance matrix of LR-HSI and the endmember matrix of HR-MSI. The tensor representation-based HSI fusion method treats the image as multidimensional data, which can better preserve the spatial–spectral structure of the image. For example, Tucker [28] proposed a fusion method based on Tucker decomposition, which assumes that HR-HSI can be estimated by the product of the core tensor and the factor matrix of the three modes, resulting in better fusion performance.

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Deep learning-based fusion algorithms have shown great potential and advantages in the field of HSI and MSI fusion due to their powerful feature extraction capabilities. They are able to automatically extract and learn image features, providing more efficient and accurate fusion results. For example, Jia et al. [29] proposed a novel multiscale spatial–spectral transformer network (MSST-Net), which combines a dual-branch architecture, a spectral transformer, a space transformer, and a self-supervised pretraining strategy to effectively improve the performance of HSI and MSI fusion. Gao et al. [30] proposed an enhanced unmixed-heuristic unsupervised network with attention-embedded degenerative learning to achieve HSI-MSI fusion. The network encoder part consists of a spatial–spectral double-stream subnetwork for modal saliency representation learning and a parameter-shared single-stream subnetwork for modal interactive representation enhancement. By introducing additional processing streams, the network is able to capture richer contextual information and cross-modal interactions, supporting more complex fusion tasks. Sun et al. [31] also proposed a transformer-based dual spatial–spectral pyramid network (DSPNet)-based HSI-MSI fusion model for three-branch cross-modal interaction. The model extracts the local spatial and spectral information of the source image through the spatial pyramid and the spectral pyramid, respectively, and obtains the HR-HSI through multiscale information fusion. In order to solve the problem of cross-modal information fusion between spatial modes and spectral modes, Zhang et al. [32] also proposed an HSI-MSI fusion model based on an interpretable spatial–spectral reconstruction network (SSR-Net), which also obtains excellent fusion performance by constructing spatial edge loss and spectral edge loss to constrain spatial and spectral distortion in the process of image fusion. Sun et al. [33] proposed a multiscale deep cross-fusion transformer (MDC-FusFormer) network for the fusion of HSI and MSI and realized the interactive fusion of spatial–spectral features through a three-branch architecture and a multiscale deep cross-fusion attention module, so as to improve the quality of the fusion image. The scarcity of training samples resulted in the recent proliferation of numerous unsupervised fusion techniques. Liu et al. [34] developed a fusion approach for HSI and MSI by using the unsupervised multiattention-guided network, UMAG-Net. On this basis, Liu et al. [35] constructed a lightweight-attention cyclic network called Circle-Net, which did not rely on labeled data for pretraining. This fusion model can directly create HR-HSI from a pair of input images without any training.

After analyzing the existing deep learning HSI and MSI fusion algorithms, we find that they have made significant progress in feature extraction and fusion, but there is a need for improvement in cross-scale integration, multilevel cross-modal interaction, and comprehensive utilization of spatial characteristics. Although MDC-FusFormer attempts to enhance the spatial–spectral feature interaction, it still has shortcomings in the efficient fusion of cross-scale and deep cross-modal information. In view of these shortcomings, inspired by the existing work, the proposed algorithm not only inherits the advantages of multibranch structure to independently process and integrate the different characteristics of HSI and MSI but also focuses on solving the problems of insufficient cross-scale information fusion and lack of multilevel cross-modal interaction. By introducing the MFF module and the PCP module, the proposed algorithm realizes the deep feature fusion from local to global and from single scale to multiscale, which effectively improves the spatial resolution and spectral fidelity of the fused image.

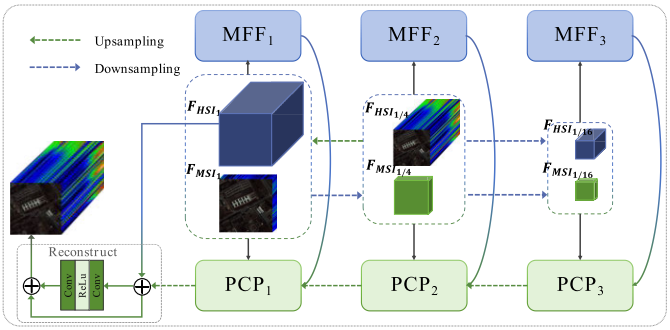


Fig. 1. Overall structure of the AMSF-Net. F_{HSI_i} and F_{MSI_i} ($i \in [1, 1/4, 1/16]$) are the different-scale HIS-MSI features obtained by upsampling/downsampling the input image. The green dotted line represents the upsampling operation, and the blue dotted line represents the downsampling operation. We performed feature extraction and fusion on the scales of 1, 1/4, and 1/16. MFF_i and PCP_i represent modules at the corresponding scales.

III. Proposed Method

Generally, HR-MSI and LR-HSI can be seen as being obtained by spectral downsampling and spatial downsampling from HR-HSI. In this article, $F_{\text{LR-HSI}} \in \mathbb{R}^{H\times W\times C}$ represents HR-HSI, where H, W, and C represent height, width, and spectral band number, respectively. HR-MSI can be represented as $F_{\mathrm{MSI}_1} \in \mathbb{R}^{H\times W\times c}$, and LR-HSI can be represented as $F_{\mathrm{HSI}_{1/4}} \in \mathbb{R}^{h\times w\times C}$, where c is the spectral band number of HR-MSI ($c \ll C$) and w and h are the width and height of LR-HSI, respectively ($h = H/r$, $w = W/r$), with r being the scaling factor, which is set to $r = 4$. The constructed AMSF-Net mainly consists of three MFFs and PCPs, as depicted in Fig. 1.

Initially, the AMSF-Net performs downsampling operations on the input images, HR-MSI and LR-HSI. To preserve more image details and features, we employ convolutional operations to downsample HR-MSI and LR-HSI, resulting in $F_{\mathrm{MSI}_{1/4}} \in \mathbb{R}^{H/4\times W/4\times c}$, $F_{\mathrm{MSI}_{1/16}} \in \mathbb{R}^{H/16\times W/16\times c}$, and $F_{\mathrm{HSI}_{1/16}} \in \mathbb{R}^{h/4\times w/4\times C}$. Meanwhile, the AMSF-Net requires an upsampling operation on LR-HSI to obtain $F_{\mathrm{HSI}_1} \in \mathbb{R}^{4h\times 4w\times C}$. Here, the bilinear interpolation method is chosen to perform the upsampling operation, as bilinear interpolation is a smoothing operation that typically does not introduce noise to the enlarged image and can improve image details and edges to a certain extent. To better extract important features from the images and enhance the precision and accuracy of feature extraction, we utilize a convolutional block attention module (CBAM) [36] after all sampling operations. The aforementioned process can be described as

$$F_{\mathrm{MSI}_{1/4}} = \mathrm{Att}(\mathrm{Down}_{\downarrow 4\times}(F_{\mathrm{MSI}_1})) \tag{1}$$

$$F_{\mathrm{MSI}_{1/16}} = \mathrm{Att}(\mathrm{Down}_{\downarrow 4\times}(F_{\mathrm{MSI}_{1/4}})) \tag{2}$$

$$F_{\mathrm{HSI}_{1/16}} = \mathrm{Att}(\mathrm{Down}_{\downarrow 4\times}(F_{\mathrm{HSI}_{1/4}})) \tag{3}$$

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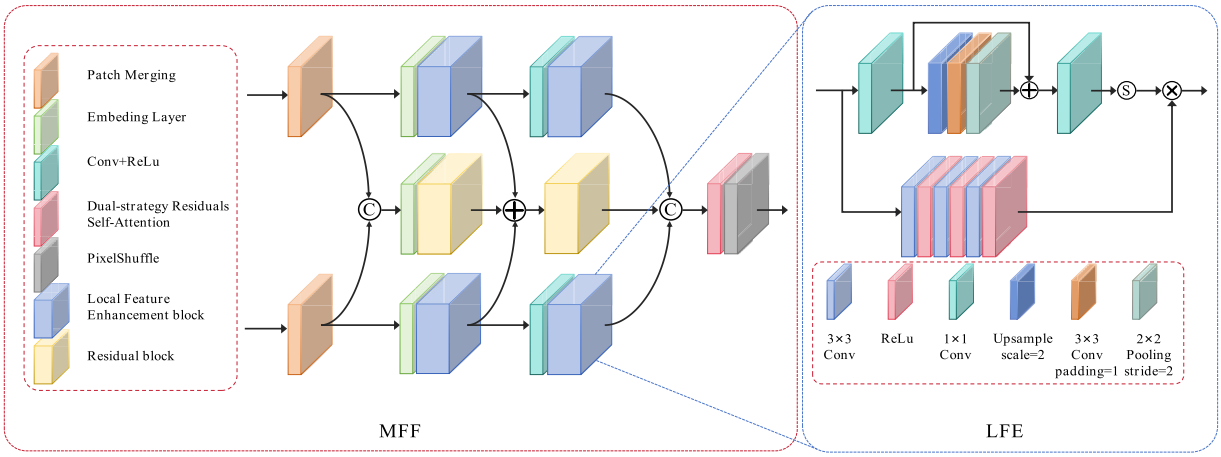


Fig. 2. Structure of MFF and LFE. The MFF module achieves information interaction between HSI and MSI at the same scale and performs feature fusion at different levels.

$$F_{\mathrm{HSI}_1} = \mathrm{Att}(\mathrm{UP}_{\uparrow 4\times}(F_{\mathrm{HSI}_{1/4}})) \tag{4}$$

where $\mathrm{Down}_{\downarrow 4\times}(\cdot)$ and $\mathrm{Up}_{\downarrow 4\times}(\cdot)$ represent the downsampling operation implemented through convolution and the upsampling operation implemented through bilinear interpolation, respectively. The parameter r denotes the sampling ratio and $\mathrm{Att}(\cdot)$ represents the mapping function of the CBAM.

Subsequently, we input the features with the same dimensional size of HSI and MSI into the MFF module together to perform feature extraction and fusion on the global features, resulting in the output features Fea_1, Fea_2, and Fea_3. The specific calculation process can be expressed as

$$\mathrm{Fea}_1 = f_{\mathrm{MFF}_1}(F_{\mathrm{MSI}_1}, F_{\mathrm{HSI}_1}) \tag{5}$$

$$\mathrm{Fea}_2 = f_{\mathrm{MFF}_2}(F_{\mathrm{MSI}_{1/4}}, F_{\mathrm{HSI}_{1/4}}) \tag{6}$$

$$\mathrm{Fea}_3 = f_{\mathrm{MFF}_3}(F_{\mathrm{MSI}_{1/16}}, F_{\mathrm{HSI}_{1/16}}) \tag{7}$$

where $f_{\mathrm{MFF}_i}(\cdot)$ represents the mapping functions of the MFF module. We fuse the spatial and spectral information of HSI and MSI through the MFF module at various levels, allowing us to maintain both high-level global structural information and low-level local detail information, such as edge textures.

Then, we adopt a multiscale feature fusion strategy, concatenating the output features Fea_i of the MFF module with their corresponding input features and feeding them into the PCP module. Through dilated convolution of different scales, horizontal and vertical features are extracted on different scale branches, and these features are input to the next branch step by step, and the output feature F_i is finally combined, which can be expressed as

$$F_3 = f_{\mathrm{PCP}_3}(\mathrm{Concat}(\mathrm{Fea}_3, F_{\mathrm{MSI}_{1/16}}, F_{\mathrm{HSI}_{1/16}})) \tag{8}$$

$$F_2 = f_{\mathrm{PCP}_2}(\mathrm{Concat}(\mathrm{Fea}_2, F_{\mathrm{MSI}_{1/4}}, F_{\mathrm{HSI}_{1/4}}, \mathrm{UP}_{\uparrow 4\times}(F_3))) \tag{9}$$

$$F_1 = f_{\mathrm{PCP}_1}(\mathrm{Concat}(\mathrm{Fea}_1, F_{\mathrm{MSI}_1}, F_{\mathrm{HSI}_1}, \mathrm{UP}_{\uparrow 4\times}(F_2))) \tag{10}$$

where $\mathrm{Concat}(\cdot)$ represents the concatenation operation along the channel dimension, $\mathrm{UP}_{\uparrow 4\times}(\cdot)$ stands for fourfold bilinear upsampling operation, $f_{\mathrm{PCP}_i}(\cdot)$ represents the mapping function of the ith layer PCP module, and F_i is the output feature of each layer of the PCP module.

To preserve the spectral and spatial features of the source images, the proposed approach uses a skip connection to concatenate the feature map F_{HSI_1} and F_1 and then inputs them into the final image reconstruction module to acquire the ultimate estimated HR-HSI $\hat{G}$, which can be expressed as

$$\hat{G} = f_{\mathrm{Rec}}(\mathrm{Concat}(F_{\mathrm{HSI}_1}, F_1)) \tag{11}$$

where $f_{\mathrm{Rec}}(\cdot)$ represents the image reconstruction module, which is composed of two consecutive convolutions and a skip connection. In the following, we shall present a comprehensive overview of the distinct compositions of the MFF and PCP.

A. MFF Module

LR-HSI and HR-MSI possess different characteristics, with LR-HSI having richer spectral information and HR-MSI containing more spatial details. In contrast to natural images, remote sensing images have a broader spatial coverage, encompassing both large-scale building outlines and small-scale local textures and details within the same scene. To effectively utilize the spectral information from HSI and the multiscale spatial information from MSI and avoid any loss of information in scenarios involving either a single stream or dual streams, we design the MFF module. This module allows for the comprehensive integration of information from both the spectral and spatial domains. Fig. 2 shows the detailed construction of the MFF module.

In this network, we use three MFF modules to receive HSI data and MSI data at three scales and obtain features from the input HSI data, MSI data, and stitched MSI and HSI data on each of the three scales. As shown in Fig. 2, MFF consists of three branches: the HSI stream branch, the MSI stream branch, and the fusion stream branch. MFF utilizes embedding layers, patch merging layers, and LFE blocks to construct the HSI stream branch and MSI stream branch. The HSI stream and MSI stream branches first use embedding layers and patch merging layers to reduce the channel and spatial dimensions of the input features to improve the computational efficiency and then extract the local features through two LFE blocks, which can be expressed as

$$F_{\mathrm{HS}_1} = f_{\mathrm{LFE}}(f_{\mathrm{PatchMerging}}(f_{\mathrm{Embeding}}(F_{\mathrm{HS}_0}))) \tag{12}$$

$$F_{\mathrm{HS}_2} = f_{\mathrm{LFE}}(\mathrm{Conv}_{1\times 1}(F_{\mathrm{HS}_1})) \tag{13}$$

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where F_{HS_0} represents the entered HSI feature, which is Z_i; $f_{\mathrm{LFE}}(\cdot)$ represents the mapping function of the LFE block; and F_{HS_1} and F_{HS_2} represent the output features of the LFE block in the HSI stream branch.

As shown in Fig. 2. The LFE block consists of two parallel branches, which can be combined with multiscale feature enhancement and residual learning for feature enhancement. A branch of the module extracts local features from the input feature map through upsampling-convolution-pooling operations and generates a dynamic attention map to weight important feature regions. The other branch of the LFE block is residual learning through deep convolution operation, which can effectively prevent the overfitting of the network while reducing the loss of input features. Finally, the LFE block realizes the enhancement of local features and the retention of overall information by multiplying the outputs of the two branches, so as to improve the model's ability to represent complex features.

The fusion stream branch takes the concatenation of F_{HS_0} and F_{MS_0} as input and extracts spatial–spectral features F_{HMS} from the fused data through a residual block (ResBlock) as follows:

$$F_{\mathrm{HMS}_1} = f_{\mathrm{RES}_1}(\mathrm{Concat}(f_{\mathrm{Embeding}}(F_{\mathrm{HS}_0}), f_{\mathrm{Embeding}}(F_{\mathrm{MS}_0}))) \tag{14}$$

where $f_{\mathrm{RES}_i}(\cdot)$ represents the mapping function of the ResBlock, which consists of a simple three-layer convolution and the residual connection. To increase the spatial–spectral information interaction, we adopt a multilevel fusion strategy to fuse the intermediate outputs of the three branches in advance and then input the next ResBlock

$$F_{\mathrm{HMS}_2} = f_{\mathrm{RES}_2}(\mathrm{Conv}_{1\times1}(F_{\mathrm{HMS}_1} + F_{\mathrm{HS}_1} + F_{\mathrm{MS}_1})). \tag{15}$$

Then, the MFF module splices the output features of the three branches along the channel dimension and sends them into the dual-strategy residuals self-attention (DRSA) block to enhance the context aggregation ability of the fused features. Finally, the MFF module is restored to the input feature size through pixel shuffle, which is

$$F_i = \mathrm{PixelShuffle}(f_{\mathrm{DRSA}}(\mathrm{Concat}(F_{\mathrm{HMS}_2}, F_{\mathrm{HS}_2}, F_{\mathrm{MS}_2}))) \tag{16}$$

where F_i represents the output feature of the ith layer MFF module and $f_{\mathrm{DRSA}}(\cdot)$ represents the mapping function of the DRSA block. Considering the matching degree between the feature size and the convolution kernel size, in order to prevent unnecessary information addition caused by too large convolution kernel, patch merging layer and pixel shuffle are not used in the third-layer MFF module.

Commonly used attention modules (such as CBAM) calculate attention weights based on scalars, which only reflect relatively important regions in the image, resulting in poor relationship modeling capabilities. Inspired by [24], a DRSA block is constructed in this article. The DRSA block supports channel interaction, which can effectively mine the potential correlation of features in all dimensions and obtain multilevel attention results through residual connection, as shown in Fig. 3. The DRSA block employs two window partitioning

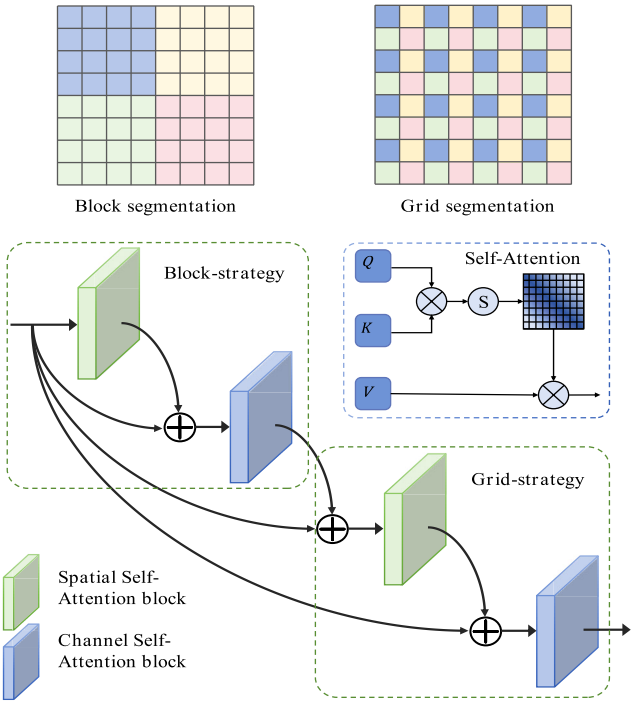


Fig. 3. Structure of the DRSA block. (Top left) Sketch map of block-based strategy and (top right) sketch map of grid-based strategy.

strategies. The first is block partitioning, as shown in Fig. 3 (top left), which splits the input feature into nonoverlapping blocks for local feature interaction. This partitioning changes the spatial dimensions of the features. The second partitioning strategy is grid partitioning, as shown in Fig. 3 (top right), which divides the input features into uniform grids for global feature interaction. The DRSA block uses the above two partitioning strategies to map the attention in turn to obtain the final aggregation results, and the attention blocks used here are based on transformer spatial self-attention and channel self-attention, which have stronger feature extraction ability and higher flexibility than simple point product attention.

B. Progressive Cross-Scale PCP Module

Relying solely on the MFF strategy of the MFF makes it difficult to fully preserve the spatial details in remote sensing images. The loss of spatial details can have a substantial impact on the quality of the fused image. Therefore, we develop a PCP module to optimize the spatial details of the feature map, and the network in this article uses three PCP modules to receive the output of the MFF module of the corresponding scale and the output of the previous PCP module on the three scales to further extract the spatial information, as shown in Fig. 1. The PCP utilizes a feature concatenation and gradual restoration strategy to acquire multiscale features from the input data. In this article, the corresponding scale PCPs are designed and attached after each scale of the MFF module to fully utilize the contextual information with a larger receptive field to restore the spatial detail information lost during the multistream branch feature extraction process.

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The PCP module first concatenates the input and output features of the same level MFF along the channel dimension. Then, it uses four dilated convolutions with different dilation rates and four BSP units to obtain features at several scales. In addition, information is supplemented by cross-scale connections between BSPs of different branches. The output of the four branches passes through different 1×1 convolutional layers to obtain feature maps of the same size, and then, the output features of each branch are added pixel-by-pixel to fuse contextual feature information of different scales, to realize the repair of the lost detail information. This process can be expressed as

$$\begin{cases} F_1 = f_{\text{BSP}}(\text{Dilate}_1(F)) \\ F_2 = f_{\text{BSP}}(\text{Concat}(\text{Dilate}_2(F), F_1)) \\ F_3 = f_{\text{BSP}}(\text{Concat}(\text{Dilate}_3(F), F_2)) \\ F_4 = f_{\text{BSP}}(\text{Concat}(\text{Dilate}_4(F), F_3)) \end{cases} \tag{17}$$

$$\text{Fea} = \text{Conv}_{3\times3}\left(\sum_{i=1}^{4}(\text{Conv}_{1\times1}(F_i))\right) \tag{18}$$

where F represents the input feature of the PCP module; F_i denotes the output feature of each branch BSP unit; Fea represents the output features of the PCP module; $\text{Dilate}_i(\cdot)$ represents the mapping function of the dilated convolution layers in each branch, whose dilation rates are detailed next; and $f_{\text{BSP}}(\cdot)$ represents the mapping function of the BSP unit.

In most CNN designs, convolutions are usually performed by using square kernels. These kernels allow the network to learn feature mappings within square windows. This approach is effective for representing most natural objects that have block-like structures. Nevertheless, for continuously distributed strip-like features, it cannot capture their linear features well, inevitably including irrelevant information from adjacent pixels and increasing unnecessary computational costs. Mei et al. [37] utilized bar-shaped convolutions to gather extensive contextual information from all four directions. Inspired by this work, we construct the BSP unit, which is seen in Fig. 4. The BSP unit first divides the features into two parts along the channel dimension and then uses the horizontal and vertical bar convolutions to extract features along the horizontal and vertical directions of the feature map, respectively. These two directional convolutions can capture features of different shapes and orientations, improving the richness and accuracy of feature representations. To capture more comprehensive feature information and better adapt to edges and textures of different directions, we add depth-wise separable convolutions after the horizontal and vertical convolutions. Then, we stitch the output features of the two branches along the channel dimension, use channel attention to learn the channel adaptive weights of the new features, and supplement the learned residual information to the original features.

To adapt to the feature size of each layer of PCP blocks, different parameters are set in different PCP layers. In $\text{PCP}_1(\cdot)$ and $\text{PCP}_2(\cdot)$, the expansion coefficient is set to [1, 2, 3, 6] and the bar convolution is set to ($k = 9$ and $p = 4$). In $\text{PCP}_3(\cdot)$, the expansion factor is set to [1, 2, 1, 2] and the bar convolution is set to ($k = 3$ and $p = 1$). Through the horizontal and vertical overlapping method, the BSP's feature extraction ability in all directions and dimensions is improved.

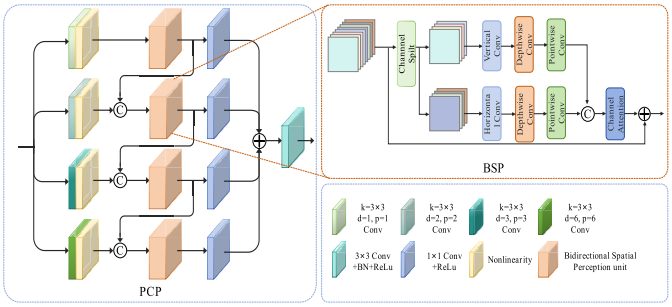


Fig. 4. Structure of the PCP module. In the PCP module, four different dilated convolutions are used to turn the features into four branches, and then, the four features are added for bidirectional spatial feature extraction step by step.

C. Loss Function

Most HSI-MSI fusion algorithms only use the mean squared error (mse) loss function during model training to calculate the average error, lacking sufficient attention to the structural and textural information. This results in improved objective evaluation metrics for the test algorithms but poor visual effects. Therefore, we utilize spatial edge loss [32], spectral edge loss [32], and mse loss to construct the overall loss function. The spatial edge loss and the spectral edge loss are constrained by calculating the spatial and spectral edges of the ground truth (GT) image and the fused result image, respectively, and calculating the mean square error of these edge information to constrain the fusion network. We can effectively narrow the distance between the two images in spatial and spectrum, thereby better ensuring that the edge details of the spatial spectrum are well preserved. The spatial edge loss L_{spat} function can effectively consider the high-frequency textural and structural information in the image, hence improving the model's capacity to retain image details. The specific loss function can be stated as

$$L_{\text{spat1}} = \frac{\sum_{k=1}^{C}\sum_{i=1}^{H-1}\sum_{j=1}^{W}\left(E_{\text{spat1}}(i,j,k) - \hat{E}_{\text{spat1}}(i,j,k)\right)^2}{2WC(H-1)} \tag{19}$$

$$L_{\text{spat2}} = \frac{\sum_{k=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W-1}\left(E_{\text{spat2}}(i,j,k) - \hat{E}_{\text{spat2}}(i,j,k)\right)^2}{2HC(W-1)} \tag{20}$$

$$L_{\text{spat}} = 0.5 * L_{\text{spat1}} + 0.5 * L_{\text{spat2}} \tag{21}$$

where L_{spat1} and L_{spat2} refer to the spatial vertical edge loss and spatial horizontal edge loss, respectively, while $E_{\text{spat1}} \in \mathbb{R}^{(H-1)\times W\times C}$ and $E_{\text{spat2}} \in \mathbb{R}^{H\times(W-1)\times C}$ represent the edge maps of the GT image G along the spatial height and width, respectively. Similarly, $\hat{E}_{\text{spat1}} \in \mathbb{R}^{(H-1)\times W\times C}$ and $\hat{E}_{\text{spat2}} \in \mathbb{R}^{H\times(W-1)\times C}$ represent the edge features of the fused image $\hat{G}$ along the spatial height and width, respectively.

The high-frequency features contained in the spectral edges are also crucial for spectral reconstruction. To maintain the

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spectral information, we incorporate a spectral edge loss function L_{spec} during model training, which is designed as

$$L_{\text{spec}} = \frac{\sum_{k=1}^{C-1}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(E_{\text{spec}}(i,j,k) - \hat{E}_{\text{spec}}(i,j,k)\right)^2}{2HW(C-1)} \tag{22}$$

where $E_{\text{spec}} \in \mathbb{R}^{H\times W\times(C-1)}$ and $\hat{E}_{\text{spec}} \in \mathbb{R}^{H\times W\times(C-1)}$ are the edge features of G and $\hat{G}$ along the spectral dimension. We use L_1 to represent spatial–spectral edge loss, that is,

$$L_1 = L_{\text{spec}} + L_{\text{spat}}. \tag{23}$$

The overall loss function L is obtained by combining the mse loss L_2 and spatial–spectral edge loss L_1, and the computation process is as follows:

$$L = \lambda L_1 + L_2 \tag{24}$$

$$L_2 = \frac{\sum_{k=1,i=1,j=1}^{C,H,W}\left(\hat{G}(i,j,k) - G(i,j,k)\right)^2}{2WHC} \tag{25}$$

where λ is the weight factor used to adjust the balance between L_1 and L_2.

IV. Experimental Setup

A. Dataset

To comprehensively assess the fusion performance of the algorithm under consideration, we opted to utilize three simulated datasets and a real-world hyperspectral dataset to test the fusion algorithm. The test dataset includes the CAVE dataset [38], the Chikusei dataset [39], the Houston dataset [40], and the WorldView-3 dataset [41]. Fig. 5 presents some training images from four datasets. In order to obtain the corresponding LR-HSI and HR-MSI, according to the widely accepted Wald protocol [42], a 5 × 5 Gaussian filter with a standard deviation of 2 is used to obfuscate the HR-HSI. Then, the fuzzy bHSI is downsampled by fourfold in the horizontal and vertical directions to obtain the corresponding LR-HSI and the HR-MSI is obtained by spectral degradation.

The CAVE dataset comprises 32 HSIs captured indoors. Each HSI has a spatial size of 512 × 512 and consists of 31 spectral bands. The images were acquired using a wavelength interval of 10 nm within 400–700 nm. We employ a dataset consisting of 20 randomly selected HSIs for training. The remaining 12 HSIs are allocated for validation and testing. HR-MSI is derived by reducing the resolution of HR-HSI using the spectral response function specific to the Nikon D700 camera.

The Chikusei dataset is HSI obtained from the Hyperspec-VNIR-CIRIS spectrometer in Chikusei, Japan. The spatial resolution of HSI is 2.5 m, with a size of 2517 × 2335, and includes 128 spectral bands spanning 363–1018 nm. When processing the HSI data, we first excluded the bands with low signal-to-noise ratio, thereby reducing the number of bands to 110. Next, we select an area of 2048 × 2048 in the HSI data center for segmentation and normalization, obtaining 16 image blocks of 512 × 512 × 110. The first ten images are designated as the training set, while the remaining six images are designated as the test set.

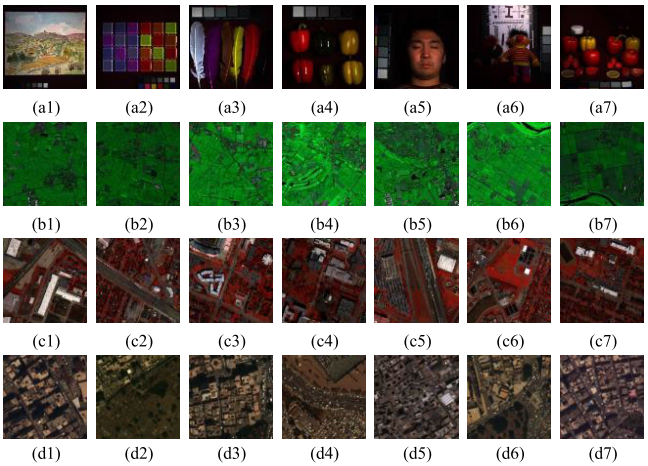


Fig. 5. Examples of pseudo-color images from the CAVE, Chikusei, and Houston datasets. (a1)–(a7) CAVE dataset with spectral bands 5, 6, and 30. (b1)–(b7) Chikusei dataset with spectral bands 20, 36, and 100. (c1)–(c7) Houston dataset with spectral bands 6, 14, and 28. (d1)–(d7) WorldView-3 dataset with spectral bands 1, 2, and 4.

The Houston dataset is a collection of remote sensing images made available by the University of Houston HSI Analysis Laboratory and the National Airborne Laser Surveying and Mapping Center of the United States. These images were published specifically for the 2018 IEEE GRSS Data Fusion Competition. The HSI data comprise 48 spectral bands, each with a spatial size of 1202 × 4172 and a spatial resolution of 1 m. This HSI data cover a spectral range from 380 to 1050 nm, with an interval of 10 nm between consecutive bands [43]. After eliminating noise and water absorption bands, we divide the HSI into 2 × 7 networks and crop them into subimages with dimensions of 512 × 512 × 46 as reference images. To create the training set, randomly select seven image blocks from the HSI. Similarly, the remaining seven image blocks are allocated for validation and testing.

The WorldView-3 dataset is mainly composed of panchromatic images and MSIs (with a number of eight bands) acquired by WV3 satellites in the visible and near-infrared spectral ranges, with spatial sampling intervals of 0.3 and 1.2 m for PAN and MSI, respectively, and a spatial resolution ratio of 4. The training set of the dataset is trimmed into small chunks for training, with the PAN size of 64 × 64 and the LR-MSI of 16 × 16, and the test set consisting of 20 pairs of PANs of size 512 × 512 and LR-MSIs of size 128 × 128.

B. Model Parameter

The AMSF-Net was trained by using the Adam optimizer for 20 000 iterative, and its formula is expressed as follows:

$$\text{lr} = \text{lr}_0 * g^{(\text{floor}(\text{epochs}/\text{step}))} \tag{26}$$

where lr_0 represents the initial learning rate, which is set to $1e^{-3}$ in this article, and lr represents the learning rate of the current training round. g is the attenuation factor, which is set to 0.95. epochs and step represents the number of training sessions and adjustment step size, which are set to 20 000 and 50. floor(·) represents the rounding function

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downward. The code of the proposed method is available at https://github.com/cvmdsp/AMSF-Net.

During the training phase, we augment the number of training samples by randomly selecting images from the associated training set and using random cropping to generate samples with pixel sizes of 64 × 64. To enrich the variety of the training data, we randomly apply flipping and rotation transformations. In addition, we adjust the batch size to 128. The presented algorithm in this article was experimented with PyTorch 11.3.0 on Python 3.8 and evaluated on MATLAB 2022a. Training and testing were conducted on a workstation, which has a Windows 11 operating system with an Intel[1] Xeon[1] Silver 4214R CPU @ 2.40 GHz and NVIDIA RTX A6000 GPU.

C. Evaluation Indexes

In addition to evaluating the performance of various fusion algorithms through subjective visual effects, we use eight popular objective indicators to comprehensively evaluate the performance of the HSI-MSI fusion algorithm on the simulated datasets. These eight objective indicators are correlation coefficient (CC), the error of relative global accuracy in synthesis (ERGAS) [44], peak signal-to-noise ratio (PSNR), root-mean-square error (RMSE), spectral angle mapper (SAM) [45], structural similarity index metric (SSIM) [46], and general universal image quality index (UIQI) [47] and mean absolute error (MoAE). For real datasets, we have chosen a nonreference metric called referenceless image spatial quality evaluator (RISQUE) [48] for evaluating the performance of various fusion algorithms.

CC is utilized to assess the degree of resemblance between two images, with a higher CC value indicating a greater similarity between them and better performance of the fusion algorithm. The PSNR of HSI is calculated as the mean value across all spectral bands. The higher the PSNR value, the smaller the difference between the fused and original images, and the more details are retained. SAM is an indicator used to estimate the spectral quality of images. The average spectral angle throughout the entire spatial domain is calculated to obtain it. The lower the SAM value, the smaller the spectral distortion, and the better the performance of the fusion algorithm. SSIM is employed to assess the structural resemblance between two pictures. A higher SSIM value indicates better preservation of the spatial structure of the image and improved performance of the fusion process. ERGAS is specifically employed to assess the overall quality of fused pictures. A lower ERGAS number indicates a higher level of spectral congruence between the fused image and the GT image, which suggests that the fusion method performs better. RMSE serves as a metric to quantify the discrepancy between the fused image and the GT image. A lower RMSE indicates the superior performance of the fusion algorithm. UIQI evaluates the quality of fused images by measuring the correlation loss, brightness distortion, and contrast distortion between the fused image and the reference image, where a value closer to 1 signifies superior performance. In HSI fusion, MoAE represents the mean value of the absolute errors across all spectral bands. A smaller MoAE indicates a lower deviation between the fused image and the GT image, indicating better fusion accuracy. RISQUE can effectively quantify the degree of image distortion and provide reliable quality assessment results by analyzing the statistical characteristics of natural scenes in images.

[1]Registered trademark.

V. Experimental Results and Analysis

To assess the effectiveness of our method, we compared AMSF-Net with seven HSI fusion algorithms, four of which are traditional: the HSI-MSI fusion approach utilizing coupled nonnegative matrix factorization (CNMF) proposed in [27], the HSI-MSI fusion approach utilizing coupled tensor double-factor (CTDF) decomposition proposed in [20], the HSI-MSI fusion approach that employs nonlocal sparse tensor factorization (NLSTF) presented in [49], which introduces tensor factorization HSI super-resolution and approximation of HSI cubes by the core tensor utilization of a dictionary based on three modes of Tucker synthesis, and the HSI-MSI fusion approach utilizing low-rank tensor sequence representation (LTTR) presented in [50], which further reconstructs HSI by grouping similar cubes into 4-D tensors and applying LTTR before the grouped 4-D tensors. In addition, there are three image fusion algorithms based on supervised deep learning, namely, the HSI-MSI fusion approach utilizing the multilevel cross transformer (MCT) proposed in [51], which realizes information complementarity through multilevel cross-modal interaction and feature aggregation reconstruction, the HSI-MSI fusion approach based on dual spatial–spectral pyramid network (DSP-Net) proposed in [31], and the HSI-MSI fusion approach employs spectral–spatial attention U-Net (SSAU) proposed in [52], which enhances the spectral and spatial information of an image with coordinate attention and efficient pyramid segmentation attention. The last comparative algorithm is based on an unsupervised deep learning image fusion algorithm, named Circle-Net proposed in [35].

A. Experimental Results of CAVE Dataset

We conducted a test on 12 images selected at random from the CAVE dataset. Table I presents the average objective assessment indicators achieved by each fusion algorithm for these 12 images. The optimal value of the objective indicators is represented in red font, while the suboptimal value is represented in blue font.

From Table I, our algorithm achieves competitive results. Among the eight objective evaluation indicators, our algorithm obtained six optimal values and two suboptimal values, with the suboptimal value only showing a difference of 0.0003 and 0.0017 compared to the optimal value. DSP-Net and Circle-Net ranked second in terms of convergence performance, achieving two optimal and four suboptimal values, respectively. The traditional method LTTR has also achieved good objective indicators, far exceeding the deep learning algorithm MCT. Our algorithm outperforms other algorithms, achieving an ERGAS index that is 24.32% lower than the

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TABLE I
QUANTITATIVE EVALUATION OF EACH FUSION ALGORITHM ON THE CAVE DATASET

Method	RMSE ↓	PSNR ↑	ERGAS ↓	SAM ↓	SSIM ↑	CC ↑	UIQI ↑	MoAE ↓
LTTR	1.3401	47.8271	1.6598	4.0177	0.9919	0.9982	0.9217	0.5743
NLSTF	2.8158	41.9385	3.1731	8.9240	0.9765	0.9936	0.8108	1.4289
CNMF	2.2110	43.7415	2.6715	8.1148	0.9745	0.9958	0.8271	1.1412
CTDF	1.5543	47.6011	1.8833	4.0180	0.9906	0.9971	0.9205	0.5997
SSAU	1.3708	46.5806	1.7404	4.6512	0.9906	0.9986	0.9031	0.7193
Circle-Net	1.2620	47.8465	1.5880	4.7833	0.9918	0.9987	0.8712	0.6000
MCT	3.7893	37.4777	3.7511	9.5254	0.9546	0.9942	0.7810	2.2583
DSP-Net	1.5775	46.4356	1.8749	4.3525	0.9935	0.9984	0.9244	0.7428
Ours	1.2193	47.9555	1.2773	4.0169	0.9932	0.9991	0.9227	0.5672

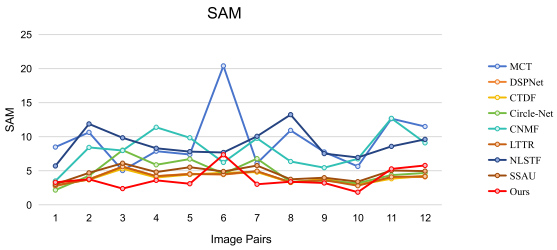


Fig. 6. SAM values of each algorithm in the CAVE test dataset.

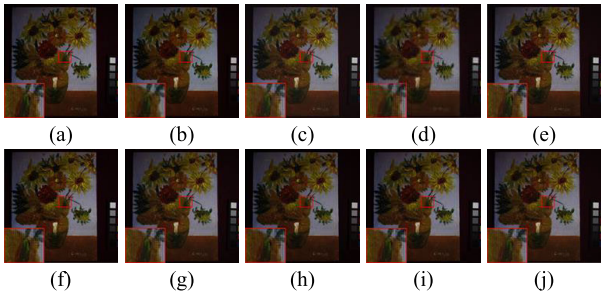


Fig. 7. Pseudo-color images of each fusion algorithm for oil painting image in the CAVE dataset. (a) GT, (b) MCT, (c) DSP-Net, (d) CTDF, (e) Circle-Net, (f) CNMF, (g) LTTR, (h) NLSTF, (i) SSAU, and (j) ours.

suboptimal value, and a PSNR that is 0.11 dB higher than the suboptimal value, indicating a significant performance improvement. This indicates that our algorithm's fusion results have smaller errors and higher image quality. This is mainly due to the multiscale image feature fusion in the constructed network being able to retain more details in images of different scales.

Considering that spectral information is an important evaluation indicator for evaluating the spectral quality and feature extraction accuracy of fused images, we present the SAM line graphs of each algorithm on 12 test images in the CAVE dataset in Fig. 6. The red lines represent the results of our algorithm, indicating that our algorithm achieved the smallest SAM value on eight images and achieved the best overall fusion effect. The complete preservation of spectral information is critical for the subsequent application of HSIs, as it is directly related to the accuracy and reliability of tasks such as image analysis, object detection, and classification.

To provide a more intuitive representation of the performance of various fusion algorithms, we randomly selected an image (named an oil painting) from the CAVE test set and converted the fusion results into the pseudo-color image for display, as shown in Fig. 7. To accentuate the image's intricate details, we enlarged the parts in red boxes that exhibit abundant color information. As depicted in Fig. 7, it is apparent that the fused images resulting from our algorithm and LTTR exhibit a closer resemblance to the GT images, whereas the fused images from other algorithms display varying degrees of blurriness and color aberration. Compared to the GT image, the fusion images of DSP-Net and MCT exhibit certain color distortion. Among them, the overall color of the fusion image in DSP-Net is dim, while in MCT, obvious black lines run through the red box area in the middle of the fusion image. There is also a black line along the left edge of the red box, which is most prominent in the white background at the top, and the overall image appears the most indistinct. The fusion result of CTDF exhibits a blurred background grid in the red box area, and the boundary between petals and leaves is not clear. The fusion image of NLSTF has a slight grid effect and heavy grid marks in the image. In the enlarged section of the CNMF fusion image, the grid lines on the left are overly conspicuous, which does not match with GT. The fusion image of Circle-Net and SSAU shows slight edge blurring on the right edge of the lemon in the enlarged area. Our algorithm and LTTR can preserve spatial and spectral information relatively completely. Our algorithm's fusion result is closest to GT.

To gain a deeper insight into the efficacy of each fusion algorithm, we have drawn error maps for each band of the fusion image and the GT image. Fig. 8 shows the error maps corresponding to the test image oil painting, which are RGB-like pseudo-color images composed of the 5th, 16th, and 30th bands. Among them, GT represents the error map corresponding to the GT image. From Fig. 8, upon inspection, the reconstruction error of the fusion images of LTTR, SSAU, and the proposed algorithm is relatively small, while the fusion image of DSP-Net produces the largest error, resulting in serious spatial distortion. The fusion images of MCT and NLSTF exhibit texture detail loss in the texture of the oil painting background. The fusion images of Circle-Net, CTDF,

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TABLE II
QUANTITATIVE EVALUATION OF EACH FUSION ALGORITHM ON THE CHIKUSEI DATASET

Method	RMSE ↓	PSNR ↑	ERGAS ↓	SAM ↓	SSIM ↑	CC ↑	UIQI ↑	MoAE ↓
LTTR	1.4020	50.5683	1.4929	1.9421	0.9861	0.9954	0.9861	0.6068
NLSTF	0.6415	54.9279	1.2087	1.0699	0.9974	0.9975	0.9913	0.2576
CNMF	1.1120	47.8378	1.6530	1.5587	0.9943	0.9974	0.9847	0.6489
CTDF	0.5859	54.8178	1.0048	0.9627	0.9975	0.9983	0.9921	0.2411
SSAU	0.6018	53.0790	0.9738	0.9969	0.9973	0.9987	0.9906	0.2848
Circle-Net	0.9590	49.2333	1.5343	1.5506	0.9948	0.9964	0.9828	0.5450
MCT	0.8243	50.5898	1.3257	1.4087	0.9960	0.9975	0.9874	0.4491
DSP-Net	0.8388	50.4378	1.3229	1.2199	0.9964	0.9974	0.9899	0.3894
Ours	0.5372	54.8444	0.8520	0.9159	0.9980	0.9988	0.9935	0.2269

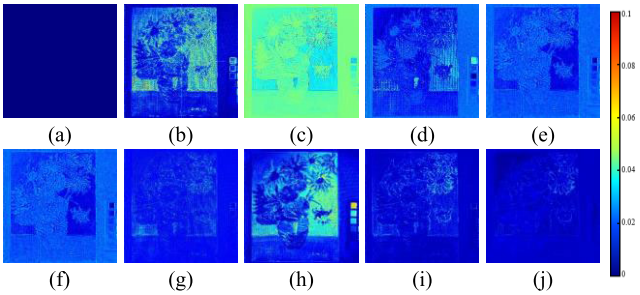


Fig. 8. Error map of each fusion algorithm for oil painting image in the CAVE dataset. (a) GT, (b) MCT, (c) DSP-Net, (d) CTDF, (e) Circle-Net, (f) CNMF, (g) LTTR, (h) NLSTF, (i) SSAU, and (j) ours.

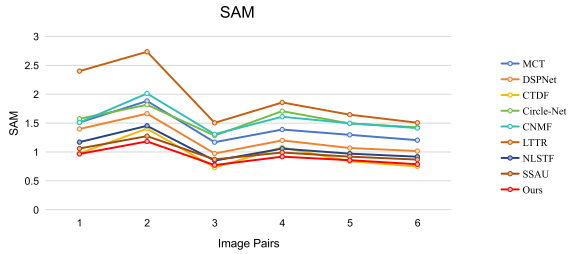


Fig. 9. SAM values of each algorithm in the Chikusei test dataset.

and CNMF exhibit a significant loss of striped detail at the edges of the vase and petals. The fusion image of LTTR and SSAU largely preserves spatial details but still lags behind our algorithm. Our algorithm can better restore the details of the image, preserving the information about the vase and the edge of the petals to the greatest extent. For the background part, many algorithms cannot balance the background information and subject information well, for example, MCT and NLSTF retain less background information, while Circle-Net and CNMF retain the subject information better, but our algorithm fully considers the spatial details of each dimension in the information extraction process and perfectly integrates the background and subject information.

B. Experimental Results of Chikusei Dataset

We randomly selected six images from the Chikusei dataset as the test set. To ensure a transparent and objective representation of the results achieved by each fusion algorithm, the average values of the objective evaluation metrics for six images have been tabulated in Table II. According to Table II, CTDF, NLSTF, and the proposed algorithm have significantly better objective evaluation metrics than other algorithms. Our algorithm achieved seven optimal values and one suboptimal value among eight objective evaluation indicators. Our algorithm is only 0.0835 dB lower than NLSTF in PSNR while improving the suboptimal values in RMSE, ERGAS, and SAM by 9.1%, 12.5%, and 5.1%, respectively, far exceeding other comparative algorithms. This fully demonstrates that our algorithm can better maintain the spatial and spectral details of the image. In addition, Fig. 9 shows the line graph plotted by each algorithm on the SAM values of six test images in the Chikusei dataset. Fig. 9 reveals a consistent trend across various images for all fusion methods. Notably, our method achieves optimal values for all images, thus reaffirming the robustness and efficiency of our approach.

For evaluating the visual effect of fused images, we chose a random image, designated as C1, from the Chikusei test set, and converted the fusion results of each algorithm into pseudo-color images, as illustrated in Fig. 10. There are many houses distributed in the C1 image, so achieving good fusion results in dense areas is a certain challenge. We enlarged and analyzed one of the small settlements (marked in red box). The fusion results of the DSP-Net, depicted in Fig. 10, exhibit low contrast and blurred details. The fusion image of NLSTF has missing contents on the left side of the red box area and insufficient information retention. The color transition in the fusion image of Circle-Net and CTDF is unnatural, presenting a clear and irregular distribution of color blocks. The fusion image of SSAU and MCT shows large color blocks and color distortion at the edges. The fusion effect of CNMF and LTTR is relatively good, but their fused images have dim colors and low contrast, resulting in spectral distortion, indicating that these algorithms have poor spectral information retention ability. Overall, ours is closest to the GT.

To better observe the effects of each fusion algorithm, we have crafted error maps for several bands, comparing the fusion image to the GT image. Fig. 11 shows the error maps corresponding to the C1 image, which are RGB-like pseudo-color images composed of the 20th, 36th, and 100th bands.

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TABLE III
QUANTITATIVE EVALUATION OF EACH FUSION ALGORITHM ON THE HOUSTON DATASET

Method	RMSE ↓	PSNR ↑	ERGAS ↓	SAM ↓	SSIM ↑	CC ↑	UIQI ↑	MoAE ↓
LTTR	5.9110	39.0946	2.1748	2.7168	0.9007	0.9827	0.9587	3.3699
NLSTF	10.9718	37.6532	3.6995	4.7391	0.8930	0.9487	0.9057	6.1731
CNMF	5.4735	39.8504	1.8817	2.2597	0.9422	0.9867	0.9698	2.9065
CTDF	4.8335	42.8774	1.6409	2.3254	0.9385	0.9899	0.9737	2.4396
SSAU	2.5091	43.7929	0.9221	1.5690	0.9795	0.9972	0.9913	1.4289
Circle-Net	4.0980	41.5380	1.4445	2.3071	0.9568	0.9923	0.9805	2.2962
MCT	3.4519	40.2992	1.3255	2.1420	0.9664	0.9944	0.9841	2.0628
DSP-Net	2.7641	44.5956	0.9638	1.5516	0.9771	0.9966	0.9910	1.4532
Ours	2.4824	44.6094	0.9091	1.5223	0.9807	0.9972	0.9916	1.3679

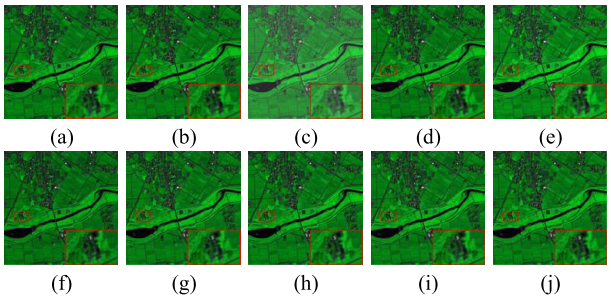


Fig. 10. Pseudo-color images of various fusion algorithms for C1 images in the Chikusei dataset. (a) GT, (b) MCT, (c) DSP-Net, (d) CTDF, (e) Circle-Net, (f) CNMF, (g) LTTR, (h) NLSTF, (i) SSAU, and (j) ours.

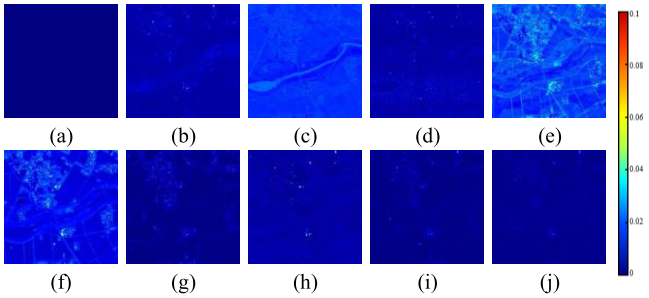


Fig. 11. Error map of each fusion algorithm for C1 images in the Chikusei dataset. (a) GT, (b) MCT, (c) DSP-Net, (d) CTDF, (e) Circle-Net, (f) CNMF, (g) LTTR, (h) NLSTF, (i) SSAU, and (j) ours.

As shown in Fig. 11, our algorithm has the cleanest error map. The error maps corresponding to other algorithms contain many bright blocks, especially the error maps of Circle-Net and CNMF, which show obvious image structures, indicating that these fusion algorithms cannot effectively extract spatial information from the source images. Through multiscale feature extraction and PCP, our algorithm can maintain the spatial texture information well, especially with the addition of spatial edge loss, we can retain the edge information of the source image.

C. Experimental Results of Houston Dataset

For the test set in the Houston dataset, to objectively demonstrate the performance of each fusion algorithm, we present the average objective evaluation indicators of each fusion algorithm on the test set in Table III. The optimal value of the objective indicators is represented in red font, while

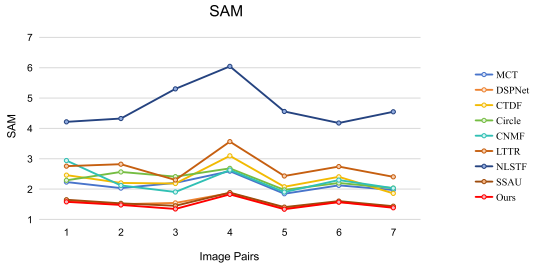


Fig. 12. SAM values of each algorithm in the Houston test dataset.

the suboptimal value is represented in blue font. According to Table III, our algorithm achieves optimal values on all objective evaluation indicators, with improvements of 1.4% and 4.3% compared to the suboptimal algorithm SSAU on objective indicators ERGAS and MoAE, and 1.9% compared to the suboptimal algorithm DSP-Net on SAM. These are sufficient evidence that our algorithm has better spatial detail information retention ability and has good spectral mapping. In addition, SSAU also achieves good results, achieving the optimal value on CC simultaneously with our algorithm while achieving suboptimal values on five objective evaluation indicators: RMSE, ERGAS, SSIM, UIQI, and MoAE. DSP-Net achieved suboptimal values on PSNR, SAM, and CC. NLSTF performs the worst on multiple objective indicators, especially RMSE, SAM, and SSIM, indicating that this algorithm generates serious spatial distortion and spectral errors, making it difficult to effectively recover image details.

Fig. 12 shows the line graph of SAM values for each algorithm on seven test images in the Houston dataset. From Fig. 12, the SAM values of NLSTF are significantly higher than other algorithms across all seven test images, while the SAM values of other algorithms are relatively close. Our algorithm has the minimum SAM values on seven test images, which also confirms the conclusion obtained in Table III that our algorithm has the smallest spectral distortion and can best preserve the spectral information of the images. This is because we have introduced a spectral edge loss function to preserve the spectral information of the image.

To gain a clearer understanding of the efficacy of each fusion algorithm, we randomly selected an image (named

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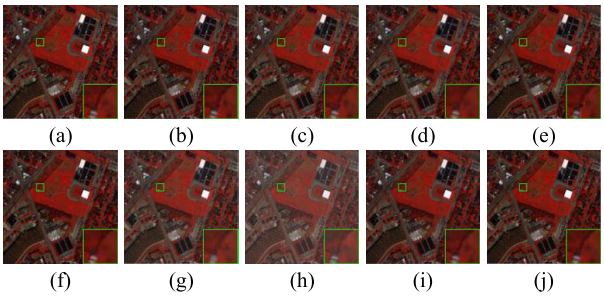


Fig. 13. Pseudo-color images of various fusion algorithms for H1 images in the Houston dataset. (a) GT, (b) MCT, (c) DSP-Net, (d) CTDF, (e) Circle-Net, (f) CNMF, (g) LTTR, (h) NLSTF, (i) SSAU, and (j) ours.

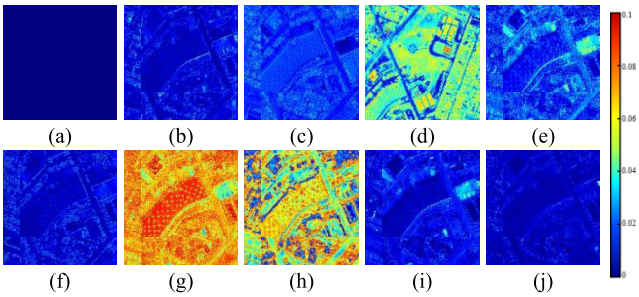


Fig. 14. Error map of each fusion algorithm for H1 images in the Houston dataset. (a) GT, (b) MCT, (c) DSP-Net, (d) CTDF, (e) Circle-Net, (f) CNMF, (g) LTTR, (h) NLSTF, (i) SSAU, and (j) ours.

H1) from the Houston test set and converted the fusion outcomes of various algorithms into a pseudo-color image for display, as shown in Fig. 13. The fusion image of LTTR is severely blurred and cannot distinguish objects within the green box. The overall color distortion of the fusion image of NLSTF indicates severe spectral information loss and incomplete retention of texture details in the image. In the fusion image of CTDF and CNMF, the black object on the right side of the selected area disappears, resulting in severe information loss. In the fused images of MCT and Circle-Net, there are artifacts in the selected regions. DSP-Net and SSAU perform relatively well in image fusion, retaining the vast majority of spatial and spectral information. However, compared to our algorithm, their fused images suffer from texture detail loss. This is because our algorithm is spatial aware in multiple dimensions, retaining more detailed information.

To better observe the effects of each fusion algorithm, we have generated error maps for several bands of the fusion image and the GT image. Fig. 14 shows the error maps corresponding to the H1 image, which are RGB-like pseudo-color images composed of bands 6, 14, and 28. Among the error maps, our method has the smallest error and the clearest error map, with only bright spots in a very small area, and the rest of the areas are well recovered. There are significant errors in the error maps of other algorithms, especially the algorithms LTTR and NLSTF, and the error of the whole image is large. Fig. 14 shows that our algorithm acquires images that are closest to GT and can largely preserve the spatial and spectral information of the source image.

TABLE IV
QUANTITATIVE EVALUATION OF EACH FUSION ALGORITHM ON THE WORLDVIEW-3 DATASET

Method	SSAU	MCT	DSP-Net	CNMF	Ours
RISQUE	53.6571	67.0849	47.6137	74.8592	**37.7480**



Fig. 15. Pseudo-color images of various fusion algorithms for W17 images in the WV3 dataset. (a) LR-HSI (b) PAN, (c) SSAU, (d) MCT, (e) DSP-Net, (f) CNMF, and (g) ours.

D. Experimental Results of Real Dataset

To verify the effect of the proposed algorithm on the real dataset, the WV3 dataset was compared with four existing methods, and the results are shown in Fig. 15. We selected the W17 image from the WorldView-3 test set and combined the 1st, 2nd, and 4th bands to create a pseudo-color image. Visualize the fusion results of different algorithms. It is obvious that the fusion result of MCT has severe color and brightness distortion. Due to the fixed input size of the MCT algorithm, during testing, we cropped it into small blocks for fusion and then stitched the fusion results together, resulting in grid-like stripes. However, the color and internal distortion of the image patch are severe, and this has nothing to do with the cropping operation. Compared to LR-HSI, there is significant ambiguity in the fusion results of SSAU. The overall fusion result of the DSP-Net fusion result contains vertical band noise, especially on the left and right edges of the image, but this is not the case with our fusion results, which preserve spatial edge detail well. The overall fusion effect of CNMF is relatively good, but there is still ambiguity. The fusion effect obtained by the proposed algorithm can well preserve the spatial detail information of PAN images, and the spectral information is also close to that of HSI images. As shown in Table IV, we score the fusion results of several algorithms using the nonreference metric RISQUE, showing the average score of all images in the test set, with lower scores representing better image quality. As can be seen from Table IV, the algorithm proposed in this article has the best score. Combined with Fig. 15, the algorithm proposed in this article has the best effect and the highest fusion quality in the spatial detail recovery of the fusion image.

E. Ablation Experiments

To enhance the validation of our algorithm's performance and thoroughly investigate the contribution of each component of the proposed model, we performed ablation studies to analyze the impact of hyperparameter choices in the loss function, the significance of the PCP module within the model, and the contribution of the MFF module to the overall performance.

1) Impact of Parameter λ: In designing the loss function, a hyperparameter λ is employed to strike a balance between the constraint ability of L_1 and L_2 during the training process.

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TABLE V
OBJECTIVE ABLATION COMPARISONS OF CHIKUSEI DATASET FOR THE DIFFERENT PARAMETERS λ

Module	RMSE ↓	PSNR ↑	ERGAS ↓	SAM ↓	SSIM ↑	CC ↑	UIQI ↑	MoAE ↓
$\lambda = 0.5$	1.0929	49.1868	1.7194	1.9180	0.9938	0.9972	0.9845	0.6813
$\lambda = 0.3$	0.8071	51.0826	1.2203	1.4170	0.9962	0.9979	0.9878	0.4574
$\lambda = 0.2$	**0.5372**	**54.8444**	**0.8520**	**0.9159**	**0.9980**	**0.9988**	**0.9935**	**0.2269**
$\lambda = 0.1$	0.9344	49.5339	1.5657	1.6301	0.9945	0.9970	0.9847	0.5820
$\lambda = 0$	1.4328	46.6444	2.0653	2.3552	0.9906	0.9956	0.9798	0.9510

TABLE VI
ABLATION STUDY OF THE PCP MODULE ON THE HOUSTON DATASET

Module	RMSE ↓	PSNR ↑	ERGAS ↓	SAM ↓	SSIM ↑	CC ↑	UIQI ↑	MoAE ↓
w-none	2.6983	43.8179	0.9569	1.6411	0.9785	0.9968	0.9910	1.4832
w-1,2,3	2.5604	44.4615	0.9334	1.5820	0.9787	0.9969	0.9914	1.4398
w-2,2,2	2.6814	43.9753	0.9587	1.6328	0.9779	0.9967	0.9909	1.4828
w-1,2,3,6	**2.4824**	**44.6094**	**0.9091**	**1.5223**	**0.9807**	**0.9972**	**0.9916**	**1.3679**
w/o BSP	2.7509	42.9993	1.0086	1.7316	0.9772	0.9966	0.9903	1.5758
w/o PCP	2.7508	43.1493	1.0032	1.7152	0.9773	0.9965	0.9902	1.5569

TABLE VII
ABLATION STUDY OF THE MFF MODULE ON THE HOUSTON DATASET

Module	RMSE ↓	PSNR ↑	ERGAS ↓	SAM ↓	SSIM ↑	CC ↑	UIQI ↑	MoAE ↓
w/o MFF	2.7867	42.9748	1.0219	1.7371	0.9768	0.9965	0.9901	1.5878
w/o fusion stream	2.7281	43.6114	0.9809	1.6517	0.9775	0.9966	0.9907	1.5144
w/o HSI/MSI stream	2.6446	43.8724	0.9515	1.6318	0.9786	0.9968	0.9911	1.4643
w/o LFE	2.7459	43.0889	1.0057	1.7135	0.9773	0.9966	0.9903	1.5651
w/o DRSA	2.7794	43.0627	1.0163	1.7423	0.9768	0.9965	0.9902	1.5797
AMSF-Net	**2.4824**	**44.6094**	**0.9091**	**1.5223**	**0.9807**	**0.9972**	**0.9916**	**1.3679**

Therefore, in ablation studies, we set λ to 0, 0.1, 0.2, 0.3, and 0.5 to verify the influence of different parameters on fusion performance. We selected the Chikusei dataset and eight evaluation metrics for experimental validation and marked the optimal values of objective evaluation metrics in bold. The results are displayed in Table V. According to Table V, when $\lambda = 0$, only using the MSE loss function, the fusion performance obtained by the model is the worst. As λ continues to increase, the fusion performance of the model reaches its optimal state when $\lambda = 0.2$. We believe that when λ is large, the loss function overly focuses on edge features and ignores the overall fusion effect, while when λ is small, the loss function overly emphasizes the overall effect and ignores the fusion of edge features. Therefore, when designing the loss function, λ is set to 0.2.

2) Effectiveness of PCP: To explore the influence of the PCP, the number of branches, and the combination of branch expansion coefficients on model experiments, four models were constructed based on the PCP blocks of the first and second layers of the model, namely: the single-branch case without using expansion convolution (represented as w-none), the expansion rate combination used for three branches is [1, 2, 3] (represented as w-1, 2, 3), [2, 2, 2] (represented as w-2, 2, 2), and the expansion rate combination used for four branches is [1, 2, 3, 6] (represented as w-1, 2, 3, 6). In addition, we also conducted ablation studies on PCP and BSP, constructing models that use single-layer convolution instead of PCP (represented as w/o PCP) and models that use double-layer convolution instead of BSP (represented as w/o BSP).

We selected the Houston dataset and eight evaluation metrics to experimentally validate the ablation model mentioned above and marked the optimal values of the evaluation metrics in bold. The results of the experiment are presented in Table VI, demonstrating that the model achieves its optimal performance when utilizing four branches with varying expansion coefficient combinations of [1, 2, 3, 6]. When the expansion coefficient combination is [1, 2, 3], the suboptimal value is obtained. It can be seen that the more branches and the larger the expansion coefficient, the larger the receptive field, and different branches can obtain multilevel features of the source images. Using the same expansion coefficient may result in branches being unable to effectively capture multilevel features, thereby affecting the performance of the network. Bearing in mind that an excessive number of branches in the network can lead to an augmentation in complexity, it can also elevate the challenges and time investment required for training and that a large expansion coefficient may cause overlap when the convolutional kernels move on the input feature map, and we adopt a four-branch structure with expansion coefficients of [1, 2, 3, 6]. Furthermore, the experiment underscored the significance of PCP and BSP, revealing that their simultaneous use leads to a notable enhancement in the model's fusion performance.

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BSP exerts a profound influence on the model's overall performance. When using single-layer 3 × 3 convolution instead of BSP, the fusion effect of the model is the worst. This may be because the multibranch dilated convolution structure leads to sparse feature maps while increasing the receptive field, while the feature information extracted by single-layer 3 × 3 convolution is insufficient to fully represent data complexity, resulting in a decrease in model performance.

3) Effectiveness of MFF: The MFF module adopts a multistream and multilevel fusion strategy, adding fusion stream branches beyond the HSI and MSI stream branches to avoid information loss in dual-stream situations, and achieving multilevel cross-modal information fusion of spectral and spatial modalities in the image. We also use dual-strategy residual self-attention blocks for channel and spatial aggregation after fusion, which is conducive to a richer and more comprehensive feature representation of the model. To demonstrate the effectiveness of MFF, LFE, and DRSA, we conducted ablation experiments on MFF. We conducted comparative experiments by constructing five models: the model with retaining only the HSI flow branch and MSI flow branch (represented as w/o fusion stream), the model with retaining only the fusion flow branch (represented as w/o HSI/MSI stream), the model using single-layer 3 × 3 convolution to replace all MFF modules in the network (represented as w/o MFF), the model with canceling all LFE blocks (represented as w/o LFE), and using single-layer 3 × 3 convolution to replace all DRSA blocks (represented as w/o DRSA). We selected the Houston dataset and eight evaluation indicators to experimentally validate the ablation model mentioned above and marked the optimal evaluation indicator values in bold. The experimental results are presented in Table VII.

Table VII indicates that the absence of the MFF module in the network leads to the poorest fusion result achieved by the model. The fusion performance of the model undergoes a gradual enhancement upon the inclusion of dual-stream branches and fusion stream branches. The fusion result experiences a significant improvement when both dual-stream and fusion stream branches are simultaneously incorporated into the network. This is because using only the dual-stream feature fusion branch will result in limited feature extraction ability, which cannot fully utilize the rich information in the data. Using only the fusion stream branch will lead to insufficient diversity in the model's feature extraction. When neither LFE nor DRSA are used in the network, the performance of the model decreases to varying degrees, which indicates that the local feature enhancement block can extract and enhance local features well, and DRSA can aggregate spatial-channel information, which is conducive to feature reconstruction. Compared with the absence of MFF, the proposed algorithm has a significant improvement effect on eight indicators, which fully proves that the constructed MFF can effectively carry out cross-modal information fusion.

VI. Conclusion

We propose an AMSF-Net for hyperspectral and MSI fusion, called AMSF-Net. AMSF-Net mainly includes MFF and PCP. MFF consists of three branches, which use the multistream fusion strategy to realize the multilevel cross-modal information fusion of HSI and MSI. PCP combines the input and output features of MFF and uses cross-scale bidirectional band convolution and deep convolution to further recover the edge features of the fused image. During training, we optimized the network by devising the comprehensive loss function of the model, which comprised spatial edge loss, spectral edge loss, and mse loss. Despite the promising results demonstrated by our method, there are still some limitations. First, the computational complexity of AMSF-Net is relatively high, which may limit its application in real-time or resource-constrained environments. Second, while our method exhibits superior fusion outcomes in terms of visual quality and objective metrics, there is still room for improvement in handling complex scenes with significant variations in spectral and spatial information.

In comparison to existing HSI-MSI fusion approaches, our method offers superior fusion outcomes. However, in our future research endeavors, we aim to address these limitations by refining and enhancing the proposed network. Specifically, we plan to explore more efficient network architectures and optimization strategies to reduce computational complexity while maintaining high fusion performance. In addition, we will investigate advanced techniques for handling complex scenes, such as incorporating deep learning-based segmentation or attention mechanisms, to further improve the robustness and accuracy of our HSI-MSI fusion approach.

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